MATH-1014 T10A

Tutorial 1

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Table of Contents

Differentiation Rules

Basic Rules:

- Power Rule: $\dfrac{d}{dx}\left[x^n\right] = nx^{n-1}$
- Exponential: $\dfrac{d}{dx}\left[e^x\right] = e^x$
- Natural Log: $\dfrac{d}{dx}\left[\ln(x)\right] = \dfrac{1}{x}$

Advanced Rules:

- Product: $(uv)' = u'v + uv'$
- Quotient: $\left(\dfrac{u}{v}\right)' = \dfrac{u'v - uv'}{v^2}$
- Chain:

$$\frac{d}{dx}\Big[f\big(g(x)\big)\Big] = f'\big(g(x)\big)\, g'(x)$$

Differentiation Rules

Problem 1: Find $\dfrac{d}{dx}\left[\dfrac{x\ln(x)}{e^x}\right]$

This problem tests:

- Product Rule
- Quotient Rule
- Natural Logarithm Differentiation
- Exponential Function Differentiation

Differentiation Rules

Problem 1: $\dfrac{d}{dx}\left[\dfrac{x\ln(x)}{e^x}\right]$

$$\begin{aligned}
\frac{d}{dx}\left(\frac{x\ln(x)}{e^x}\right) &= \frac{(x\ln x)'\,e^x - (e^x)'\,(x\ln x)}{(e^x)^2} \\
&= \frac{(x\ln x)' - x\ln x}{e^x} \\
&= \frac{\left(\ln x + x\cdot\frac{1}{x}\right) - x\ln x}{e^x} \\
&= \frac{(1-x)\ln x + 1}{e^x}.
\end{aligned}$$

Integration Techniques

- **Basic Integrals:**

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C$$

$$\int e^x \, dx = e^x + C$$

$$\int \frac{1}{x} \, dx = \ln|x| + C$$

- **Integration by Substitution:**

$$\int f(g(x))\, g'(x)\, dx \;=\; \int f(u)\, du$$

- **Integration by Parts:**

$$\int u\, dv \;=\; uv \;-\; \int v\, du$$

Problem 2: Evaluate $\int \frac{\ln(x)}{x}\sqrt{1+\ln^2(x)}\,dx$

This problem tests:

- Integration by Substitution
- Recognition of Complex Patterns
- Chain Rule Understanding

Problem 2:

Let $u(x) = \ln x$

$$du = \frac{1}{x}\,dx$$

$$\begin{aligned}
\int \frac{\ln x}{x}\sqrt{1+\ln^2 x}\,dx &= \int \frac{u(x)}{x}\sqrt{1+u^2(x)}\,dx \\
&= \int u\sqrt{1+u^2}\,du \\
&= \int u(1+u^2)^{1/2}\,du \\
&= \frac{1}{3}(1+u^2)^{3/2} + C \\
&= \frac{1}{3}\left(1+\ln^2 x\right)^{3/2} + C.
\end{aligned}$$

Problem 3
If $f(x) = 4\ln(\ln(x))$, find $f'(x)$.

Problem 4
Let $y = \tan^{-1}\left(\sqrt{3x^2 - 1}\right)$. Then $\frac{dy}{dx} =$

Identify the structure of the composite function first!

Solutions

Problem 3

If $f(x) = 4\ln(\ln x)$, find $f'(x)$.

$$\begin{aligned} \text{Let } g(x) &= \ln x, \qquad f(x) = 4\ln(g(x)). \\ f'(x) &= 4 \cdot \frac{1}{g(x)}\, g'(x) \\ &= 4 \cdot \frac{1}{\ln x} \cdot \frac{1}{x} = \frac{4}{x\ln x}. \end{aligned}$$

Problem 4

Let $y = \tan^{-1}\left(\sqrt{3x^2-1}\right)$. Find $\frac{dy}{dx}$.

$$\begin{aligned} y &= \arctan u, \qquad u = \sqrt{3x^2-1}. \\ y' &= \frac{1}{1+u^2}\, u' \\ &= \frac{1}{1+(3x^2-1)} \cdot \frac{6x}{2\sqrt{3x^2-1}} = \frac{1}{x\sqrt{3x^2-1}}. \end{aligned}$$

Integration by part

Problem 5
Evaluate $\displaystyle\int \frac{x}{\sqrt{x+5}}\,dx$.

Integration by substitution

Problem 6
Evaluate $\displaystyle\int \frac{\cos(t)}{(5\sin(t)+12)^2}\,dt$.

Solutions

Problem 5

Evaluate $\displaystyle\int \frac{x}{\sqrt{x+5}}\,dx$.

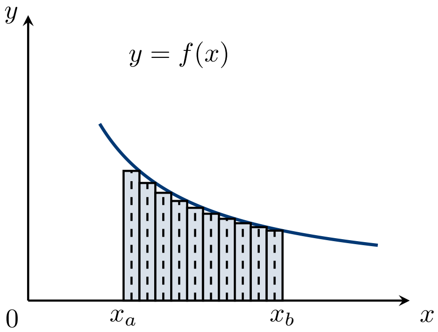
$$\int \frac{x}{\sqrt{x+5}}\,dx = 2\int x\,d\sqrt{x+5} = 2\left(x\sqrt{x+5} - \int \sqrt{x+5}\,dx\right)$$
$$= 2\left(x\sqrt{x+5} - \frac{2}{3}(x+5)^{3/2}\right) + C.$$

Problem 6

Evaluate $\displaystyle\int \frac{\cos(t)}{(5\sin(t)+12)^2}\,dt$. Let $u = \sin t, du = \cos t\,dt$

$$\int \frac{\cos t}{(5\sin t+12)^2}\,dt = \int \frac{du}{(5u+12)^2}$$
$$= -\frac{1}{5(5u+12)} + C = -\frac{1}{5(5\sin t+12)} + C.$$

Definition of Definite Integral



$$\int_{x_a}^{x_b} f(x)\,dx$$

$$\text{Area} \approx \sum_{i=1}^{n} f(x_i^*)\,\Delta x_i$$

$$\text{Area} = \lim_{\max \Delta x_i \to 0} \sum_{i=1}^{n} f(x_i^*)\,\Delta x_i$$

Riemann sum!

Fundamental Theorem of Calculus

Part 1: If $F'(x) = f(x)$, then

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

Part 2: For continuous f,

$$\frac{d}{dx}\int_a^x f(t)\,dt = f(x).$$

Problem 7 $F(x) = \displaystyle\int_0^x \sin(t^2)\,dt.$ Find $F'(x)$.

Problem 8

Evaluate the integral $\int_0^4 (2e^x + 8\cos x)\, dx$.

Try to find the antiderivative first!

Problem 9

Suppose that $F(x) = \int_1^x f(t)\, dt$, where

$$f(t) = \int_1^{t^2} \frac{\sqrt{6+u^6}}{u}\, du.$$

Find $F''(2)$.

Solutions

Problem 8

Evaluate $\int_0^4 (2e^x + 8\cos x)\; dx$.

$$\begin{aligned}\int_0^4 (2e^x + 8\cos x)\; dx &= \left[\, 2e^x + 8\sin x \,\right]_0^4 \\ &= 2e^4 + 8\sin 4 - 2.\end{aligned}$$

Problem 9

Given $F(x) = \int_1^x f(t)\, dt$ and $f(t) = \int_1^{t^2} \frac{\sqrt{6+u^6}}{u}\, du$, find $F''(2)$.

$$F'(x) = f(x) \quad \Rightarrow \quad F''(x) = f'(x).$$

$$f'(t) = \frac{\sqrt{6+(t^2)^6}}{t^2} \cdot \frac{d}{dt}(t^2) = \frac{\sqrt{6+t^{12}}}{t^2} \cdot 2t = \frac{2\sqrt{6+t^{12}}}{t}.$$

$$F''(2) = f'(2) = \frac{2\sqrt{6+2^{12}}}{2} = \sqrt{6+4096} = \sqrt{4102}.$$

Problem 10

Evaluate the following limit by first recognizing the sum as a Riemann sum for a function defined on $[0, 1]$:

$$\lim_{n\to\infty} \frac{2}{n}\left(\sqrt{\frac{1}{n}} + \sqrt{\frac{2}{n}} + \sqrt{\frac{3}{n}} + \cdots + \sqrt{\frac{n}{n}}\right).$$

Recognize it as a Riemann sum on $[0, 1]$!

Solutions

Problem 10

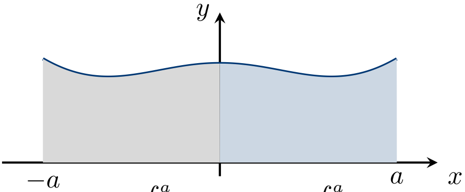
$$\lim_{n\to\infty} \frac{2}{n} \sum_{i=1}^{n} \sqrt{\frac{i}{n}} = \lim_{n\to\infty} 2 \sum_{i=1}^{n} \sqrt{\frac{i}{n}} \cdot \frac{1}{n}$$

$$= 2 \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\,\Delta x,$$

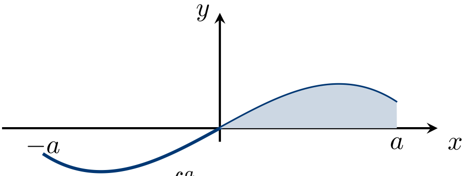
$$\Delta x = \frac{1}{n}, \quad x_i = \frac{i}{n} \in [0,1], \quad f(x) = \sqrt{x}$$

$$= 2 \int_0^1 \sqrt{x}\,dx = 2 \left[\frac{2}{3} x^{3/2}\right]_0^1 = \frac{4}{3}.$$

Symmetry



(a) f even, $\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$



(b) f odd, $\int_{-a}^{a} f(x)\,dx = 0$

Note

- Even: $f(-x) = f(x)$
- Odd: $f(-x) = -f(x)$
- Use symmetry to simplify integrals on $[-a, a]$.

Problem 11 Calculate $I = \int_{-1}^{1} \left(\ln x + \sqrt{1+x^2} + 2x + e^{-x} \right) dx.$

Solution

Problem

$$\begin{aligned} u(x) &= \ln\left(x + \sqrt{1+x^2}\right). \\ u(-x) + u(x) &= \ln\left(-x + \sqrt{1+x^2}\right) + \ln\left(x + \sqrt{1+x^2}\right) \\ &= \ln\left(\left(\sqrt{1+x^2} - x\right)\left(\sqrt{1+x^2} + x\right)\right) = \ln(1) = 0. \end{aligned}$$

Hence $u(x)$ is odd, so $\int_{-1}^{1} u(x)\,dx = 0$. Therefore

$$\begin{aligned} I &= \int_{-1}^{1} \left(u(x) + (2x+1)e^{-x}\right) dx = \int_{-1}^{1} (2x+1)e^{-x}\,dx \\ &= \int_{-1}^{1} (2x+1)\,d(e^{-x}) = -\left[(2x+1)e^{-x}\right]_{-1}^{1} + \int_{-1}^{1} 2e^{-x}\,dx \\ &= -\left(3e^{-1} - (-1)e^{1}\right) + \left[-2e^{-x}\right]_{-1}^{1} = e - \frac{5}{e}. \end{aligned}$$

THANK YOU!